IoT with Applications — Video Lecture Series Outline

5 Units × 3 Hours Each (180 min/unit) — Headings & Sub-headings for Scripting

Prepared for faculty use on a personal/academic website

UNIT I: Introduction to IoT — (180 min)

1. Course Introduction & Why IoT Matters — *10 min*

- Welcome, instructor introduction, and course context
- Roadmap: what the 5 units will cover
- Motivation: everyday IoT examples students already encounter

2. What is IoT? — *20 min*

- Definition and characteristics of IoT
- Evolution of IoT (from RFID tagging to today's AI-driven IoT)

3. IoT vs Related Concepts — *20 min*

- IoT vs Internet
- IoT vs M2M (Machine-to-Machine)
- IoT vs Cyber-Physical Systems

4. Building Blocks of IoT — *20 min*

- Components of IoT (devices, connectivity, data processing, user interface)
- IoT enabling technologies (sensors, connectivity, cloud, big data, AI)

5. IoT Architecture — *35 min*

- Three-layer architecture
- Five-layer architecture
- Perception layer (deep dive)
- Network layer (deep dive)
- Processing/Cloud layer (deep dive)
- Application layer (deep dive)
- Edge and Fog computing (introductory overview)

6. IoT Devices Explained — *35 min*

- **Sensors (types, examples)**
- **Actuators**
- **Microcontrollers**
- **Embedded systems**
- **RFID, NFC, GPS**
- **Smart devices**

7. The IoT Data Journey — *30 min*

- **Data acquisition**
- **Data transmission**
- **Data storage**
- **Data analytics**
- **Real-time monitoring**
- **IoT dashboards**

8. Unit Recap & Self-Assessment — *10 min*

- **Key-term rapid recap**
- **5-question self-assessment for students**
- **Preview of Unit II**

Unit I Total: 180 min

UNIT II: IoT Communication and Networking — (180 min)

1. Introduction: Why Communication Is the Backbone of IoT — *10 min*

- **Recap of Unit I architecture layers**
- **Framing: short-range vs long-range vs protocols vs networking concepts**

2. Short-Range Communication Technologies — *40 min*

- **Bluetooth / BLE**
- **Zigbee**
- **RFID (recap + communication angle)**

- **NFC (recap + communication angle)**
- **Wi-Fi**

3. Long-Range Communication Technologies — *35 min*

- **LoRaWAN**
- **NB-IoT**
- **LTE-M**
- **4G/5G for IoT**

4. IoT Communication Protocols — *45 min*

- **MQTT (deep dive: publish-subscribe model, broker, topics)**
- **CoAP**
- **HTTP/HTTPS**
- **AMQP**
- **WebSockets**

5. IoT Networking Concepts — *40 min*

- **IPv4 and IPv6 in IoT**
- **6LoWPAN**
- **Wireless Sensor Networks (WSN)**
- **Gateways**
- **Addressing and routing**

6. Comparison Table & Recap — *10 min*

- **Side-by-side comparison: range, power, bandwidth, use-case**
- **Self-assessment questions + preview of Unit III**

Unit II Total: 180 min

UNIT III: IoT Platforms, Edge and Cloud — (180 min)

1. Introduction — *10 min*

- **Why hardware + cloud + edge + AI matter together**
- **Where this unit fits in the overall IoT pipeline**

2. IoT Hardware Platforms — *40 min*

- **Arduino**
- **ESP32**
- **Raspberry Pi**
- **NodeMCU**
- **Sensors and modules overview (how they connect to boards)**

3. Cloud Computing for IoT — *35 min*

- **IoT cloud architecture**
- **Device management**
- **Data storage in the cloud**
- **Cloud analytics**
- **IoT dashboards (cloud-side)**

4. Edge Computing — *35 min*

- **Need for edge computing**
- **Edge vs Cloud (trade-offs)**
- **Real-time processing**
- **Edge AI**
- **Fog computing**

5. IoT + AI — *45 min*

- **Machine learning with sensor data**
- **Anomaly detection**
- **Predictive maintenance**
- **Computer vision in IoT**
- **Intelligent IoT systems**
- **Walkthrough: full pipeline — Sensor → ESP32 → Edge AI → Cloud → Dashboard → Actuator**

6. Recap & Self-Assessment — *15 min*

- **Platform comparison recap (Arduino vs ESP32 vs Pi vs NodeMCU)**
- **Edge vs Cloud decision-making questions**

- **Preview of Unit IV**

Unit III Total: 180 min

UNIT IV: IoT Security and Data Analytics — (180 min)

1. Introduction: Why IoT Security Matters — *10 min*

- **Real-world IoT breach examples (context-setting)**
- **Framing: security requirements → threats → analytics → blockchain**

2. IoT Security Fundamentals — *40 min*

- **Security requirements**
- **Authentication**
- **Authorization**
- **Encryption**
- **Secure communication**
- **Device security**
- **Network security**
- **Cloud security**

3. IoT Threats & Attacks — *30 min*

- **Device attacks**
- **Eavesdropping**
- **Spoofing**
- **Denial-of-Service (DoS) attacks**
- **Malware**
- **Data leakage**

4. IoT Data Analytics — *45 min*

- **Data preprocessing**
- **Time-series data**
- **Visualization**
- **Anomaly detection**

- **Predictive analytics**
- **AI/ML-based IoT analytics**

5. Blockchain + IoT — *35 min*

- **Decentralized IoT**
- **Smart contracts**
- **Data integrity**
- **IoT identity management**

6. Recap & Self-Assessment — *20 min*

- **Security checklist recap**
- **Analytics pipeline recap**
- **Short self-assessment questions + preview of Unit V**

Unit IV Total: 180 min

UNIT V: IoT Applications — (180 min)

1. Introduction to Unit V — *10 min*

- **How every domain below reuses the same pipeline: Sensors → Gateway → Cloud → AI → Action**

2. Smart Agriculture — *25 min*

- **Pipeline: Sensors → IoT Gateway → Cloud → AI → Irrigation**
- **Soil moisture monitoring**
- **Automatic irrigation**
- **Crop monitoring & disease detection**
- **Weather monitoring**
- **Smart fertilizer management**

3. Healthcare IoT — *25 min*

- **Remote patient monitoring**
- **ECG monitoring**
- **SpO_2 monitoring**
- **Temperature monitoring**

- **Smart medical devices**
- **Elderly-care systems & emergency alerts**

4. Industrial IoT (IIoT) — *25 min*

- **Machine monitoring**
- **Predictive maintenance**
- **Asset tracking**
- **Industrial automation**
- **Digital twins**
- **Quality monitoring**

5. Smart City — *25 min*

- **Smart traffic management**
- **Smart parking**
- **Smart streetlights**
- **Smart waste management**
- **Air-quality monitoring**
- **Water management**

6. Smart Energy & Smart Transportation — *30 min*

- **Smart meters, smart grid, demand prediction**
- **Renewable-energy monitoring, building energy management**
- **Vehicle tracking, fleet management**
- **Intelligent traffic systems, connected vehicles, accident detection**

7. Smart Home — *20 min*

- **Smart lighting**
- **Smart security**
- **Smart appliances**
- **Energy management**
- **Voice-controlled devices**
- **Intrusion detection**

8. Wrap-Up: The Future of IoT & Course Conclusion — *20 min*

- **IoT 1.0 → 2.0 → 3.0 → 4.0 evolution (Connected Things → Cloud-connected → AI-enabled → Edge AI + 5G + Digital Twins)**
- **Suggested mini-project ideas (Smart Campus, Agriculture, Healthcare, Energy, Predictive Maintenance)**
- **Final recap of all 5 units**
- **Closing remarks and pointer to the accompanying lab series**

Unit V Total: 180 min